Higher National Diploma in Engineering (Mechanical/Civil)

Higher National Diploma in Building Service Engineering

First Year, First Semester Examination – 2022

CE-1102-Fluid Mechanics

Answer any four questions only. **2hours**

Question 1 (Q1)

a) Define the following terms related to fluid with relevant equations: i) Density ii) Specific weight iii) Specific volume iv) Specific gravity v) Pascal's law (05 marks)

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b) The diameters of the small piston and the large piston of a hydraulic jack are **4 cm** and **12 cm** respectively. A force of **95 N** was applied to the small piston and the large piston moved upward without accelerating. The density of the liquid in the jack is $900\ \mathrm{kg/m^3}$. Find the load lifted by the large piston when: i) Pistons are at the same level (06 marks) ii) Small piston is **45 cm** above the large piston (06 marks)

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c) i) Write the **equation of state of a perfect gas** (03 marks) ii) A mass of air, at a pressure of **220 kPa** and a temperature of **320 K**, occupies a volume of $3.2\ \mathrm{m^3}$. Determine the **density** of the air and its **mass** (05 marks).

Question 2 (Q2)

a) Derive expressions for **Total pressure force** and distance to the **center of pressure** from the free surface of a liquid for an inclined plane surface submerged in a liquid. (10 marks)

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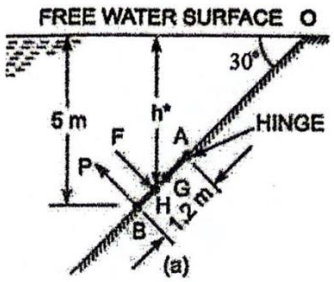
b) An inclined rectangular sluice gate AB ($1.2\ \mathrm{m} \times 4\ \mathrm{m}$ in size) is installed to control the discharge of water. The end A is hinged. Determine: i) **Total Pressure force (F)** applied at H (05 marks) ii) **Vertical distance (h^*)** from the free surface to the total pressure force (05 marks) iii) Determine the **force normal (P)** to the gate applied at B to open it (05 marks)



Question 3 (Q3)

a) Explain the following terms in brief: i) **Archimedes principal** (02 marks) ii) **Center of buoyancy** (02 marks)

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b) Briefly explain the following terms related to a floating body with suitable sketches: i) **Stable Equilibrium** (03 marks) ii) **Unstable Equilibrium** (03 marks) iii) **Neutral Equilibrium** (03 marks)

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c) A jet of water of diameter **10 cm** strikes a flat plate normally with a velocity of $\mathbf{12\ m/s}$. The plate is moving with a velocity of $\mathbf{5\ m/s}$ in the direction of the jet and away from the jet. Find: i) The **force exerted** by the jet on the plate (06 marks) ii) **Work done** by the jet on the plate per second (03 marks) iii) **Efficiency** of the jet (03 marks)

Question 4 (Q4)

a) i) State what is **dynamic viscosity** with relevant equations (02 marks). ii) State what is **kinematic viscosity** with relevant equations (02 marks). iii) Categorize the **different types of fluids** based on viscosity and briefly explain their viscosity behavior (05 marks).

b) A cubic block weight **30 kg** and having a **40 cm** edge is allowed to slide down on an incline plane surface making an angle of 30° with the horizontal on which there is a thin oil film having a viscosity of $\mathbf{0.22 \times 10^{-3}\ N \cdot s/m^2}$. What is the **terminal velocity**? (07 marks)

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c) Briefly explain the types of **turbines** used for power generation (09 marks).

Question 5 (Q5)

a) i) Briefly Explain the **continuity equation** (03 marks). ii) Briefly Explain **Bernoulli's equation** (03 marks).

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b) A horizontal **venturi meter** with inlet and throat diameters of **30 cm** and **15 cm** respectively is used to measure the flow of water. The reading of the differential manometer connected to the inlet and throat is **25 cm of mercury**. Determine the **rate of flow**. Take $\mathbf{C_d = 0.96}$ and the specific gravity of mercury is **13.6** (12 marks).

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c) Write equations for the following: i) **Discharge (flow rate) over a rectangular notch** (02 marks) ii) **Discharge (flow rate) over a triangular notch** (02 marks)

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d) Write **advantages of Triangular notch over rectangular notch** (03 marks).